Arab Academy for Science, Technology and Maritime Transportation

Fluid Mechanics 1 – ME 362

Dr. Aly Hassan Elbatran

Assistant Professor

a.elbatran@aast.edu

aly_marine@hotmail.com

Course Assistant Lecturer:

Eng. Islam

Fluid Mechanics 1 – ME 362

Lecture 2:

Physical Properties of Fluids

Measures of Fluid Mass and Weight

Density of a fluid, ρ (rho), is the amount of mass per unit volume of a substance:

$$\rho = m / V$$

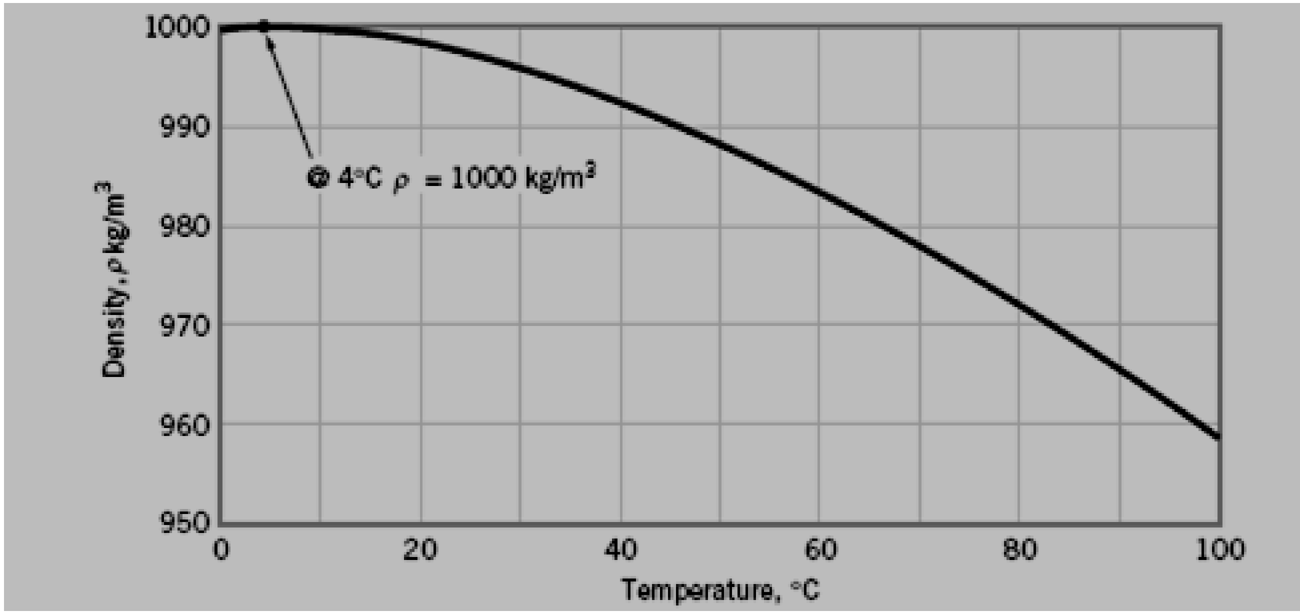
- For liquids, weak function of temperature and pressure

  Density in liquids is nearly constant due to (**slightly affected by changes in *temperature* and *pressure***)

- For gases: strong function of T and P ➡ $\rho = \rho(P, T)$

  Density is highly variable in gases and increases nearly proportionally to the pressure level

Density of water as a function of temperature



Units: kg/m^3

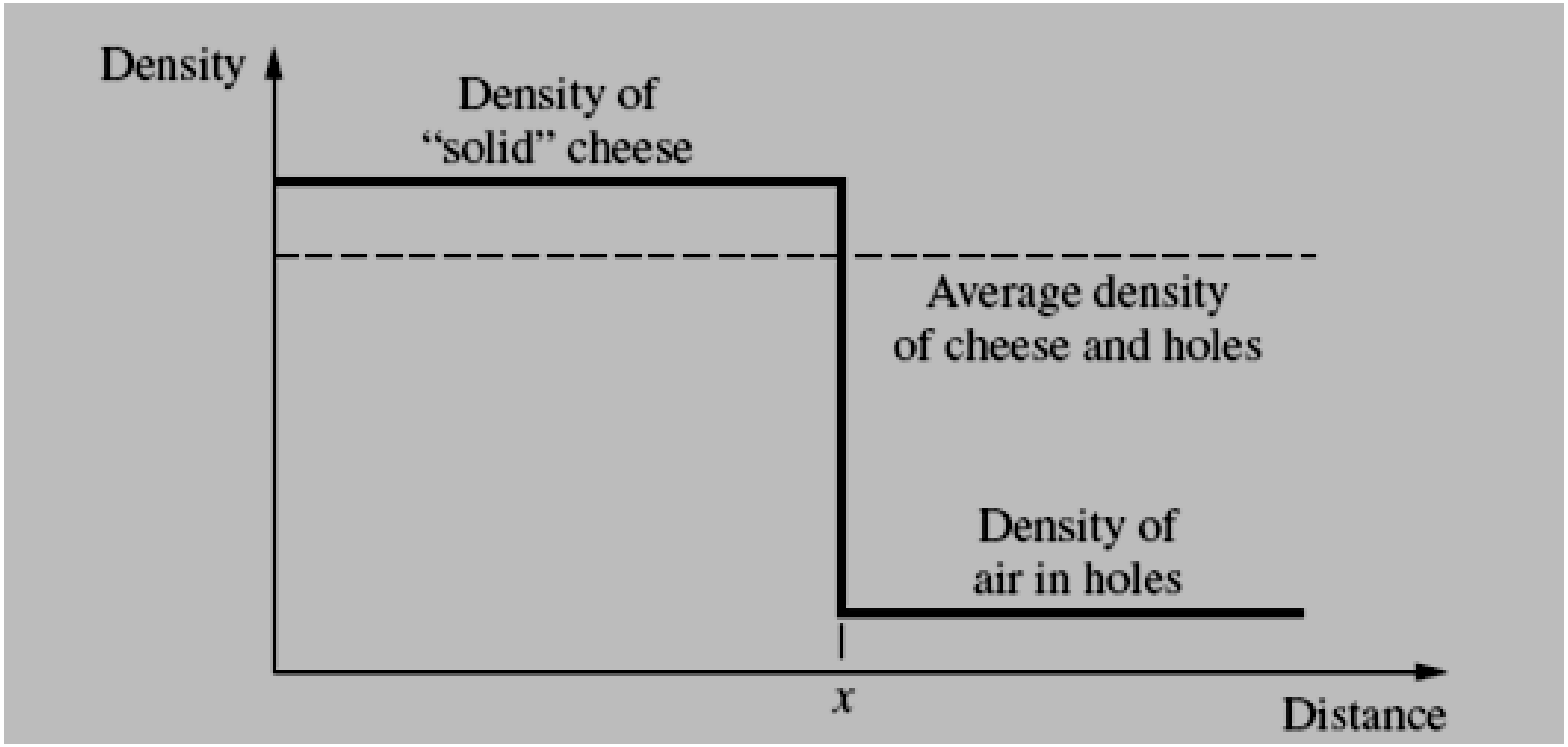
Typical values:

Water = 1000 kg/m^3;

Air = 1.23 kg/m^3;

In general, liquids are about three orders of magnitude more dense than gases at atmospheric pressure.

Density $\rho = \dfrac{m}{V}$



The density of Swiss chees is not uniform from point to point, but has local point densities an average density.

Specific Weight γ

The **specific weight of a fluid,** (also known as the **unit weight**) is the **weight** per **unit** volume. The symbol of **specific weight** is γ (the Greek letter Gamma).

$$(\gamma = \rho \, \mathbf{g})$$

Where:

g is the local acceleration of gravity. Just as density is used to characterize the mass of a fluid system, the specific weight is used to characterize the weight of the system.

Density and Specific Weight are Simply Related by Gravity

$$W = m \, \mathbf{g} \qquad \& \qquad m = W / \, \mathbf{g}$$

In the **BG system**, γ has units of Ibf/ft^3 and in **SI the units** are N/m^3. under conditions of standard gravity (**g** = 32.174 ft/s^2 = 9.807 m/s^2), water at 20 **°C** has a specific weight of **62.4 Ibf/ft^3 and 9.8 kN/m^3.**

Specific Gravity: SG

Specific gravity, denoted by **SG**, is defined as the ratio of a fluid density to a standard reference fluid density, water density (for liquids), and air density (for gases).

Usually the specified temperature is taken as 4 **°C** (39.2 **°**F), and at this temperature the density of water is 1.94 slugs/ft^3 or 998 kg/m^3.

Specific gravity is expressed as:

$$SG_{gas} = \frac{\rho_{gas}}{\rho_{air}} = \frac{\rho_{gas}}{1.205\ \text{kg/m}^3}$$

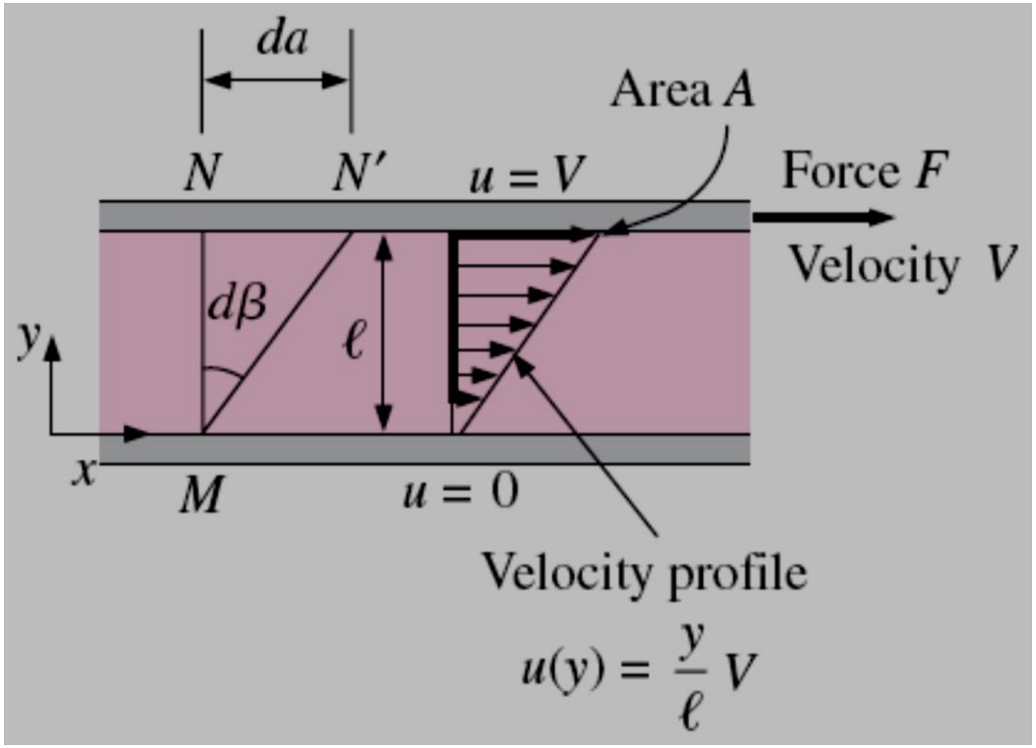
$$SG_{liquid} = \frac{\rho_{liquid}}{\rho_{water}} = \frac{\rho_{liquid}}{998\ \text{kg/m}^3}$$

Note that: *density of water at 4* °C *usually taken as (1000 kg/m^3)*

- The properties of density and specific weight are measures of the “***heaviness***” of a fluid. It is clear, however, that these properties are not sufficient to uniquely characterize how fluids behave since two fluids (such as water and oil) can have approximately the same value of density but behave quite differently when flowing.
- Apparently, some additional property is needed to describe the “**fluidity**” of the fluid.

Viscosity

- Viscosity: A property that represents the internal resistance of a fluid to motion or the "fluidity".
- The viscosity of a fluid is a measure of its "*resistance to deformation*."
- Viscosity is due to the internal frictional force that develops between different layers of fluids as they are forced to move relative to each other. ***Viscosity can then lead to energy loss***.



The behavior of a fluid in laminar flow between two parallel plates when the upper plate moves with a constant velocity.

$$\tau = \frac{F}{A}$$

$$\frac{u(y)}{y} = \frac{V}{\ell} \quad \text{and} \quad \frac{du}{dy} = \frac{V}{\ell}$$

$$d\beta \approx \tan d\beta = \frac{da}{\ell} = \frac{V\,dt}{\ell} = \frac{du}{dy}dt$$

$$\frac{d\beta}{dt} = \frac{du}{dy}$$

Newtonian fluids: Fluids for which the rate of deformation is proportional to the shear stress.

$$\tau \propto \frac{d(d\beta)}{dt} \quad \text{or} \quad \tau \propto \frac{du}{dy}$$

$$\tau = \mu \frac{du}{dy} \qquad (\text{N/m}^2)$$

Shear stress

(μ) is the Dynamic Viscosity

Shear force

$$F = \tau A = \mu A \frac{du}{dy} \qquad (\text{N})$$

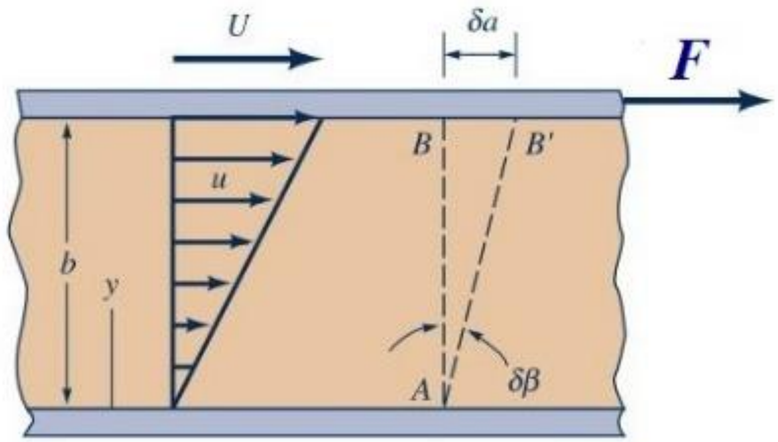
μ Coefficient of Viscosity

Dynamic (Absolute) Viscosity

$\text{kg/m} \cdot \text{s}$ or $\text{N} \cdot \text{s/m}^2$ or $\text{Pa} \cdot \text{s}$

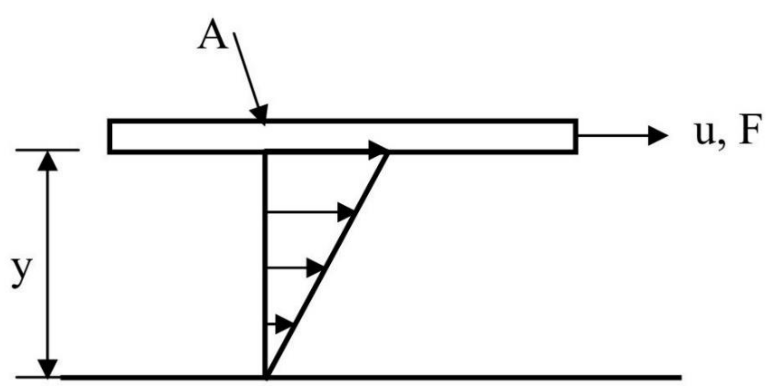
1. Flat Plate

$$\tau \propto \frac{d(d\beta)}{dt} \quad \text{or} \quad \tau \propto \frac{du}{dy}$$



$$\tau = \mu \frac{du}{dy}$$

$$Hence\ \tau = \frac{F}{A}; \qquad So: \quad F = \mu A \frac{du}{dy}$$

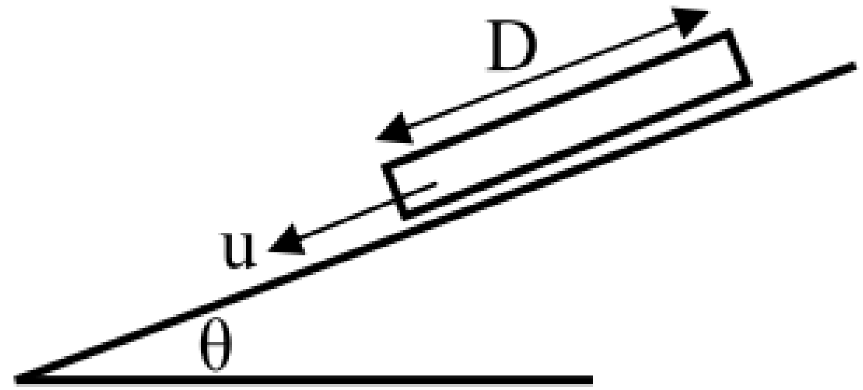


While

τ	: Shear Stress	**N/m²**
μ	: Dynamic Viscosity	**N·s/m²**
$\frac{du}{dy}$	: Rate of the shear strain	**s⁻¹**
F	: Viscous force	**N**

2. Sliding Disk

$\tau \propto \frac{d(d\beta)}{dt}$ or $\tau \propto \frac{du}{dy}$



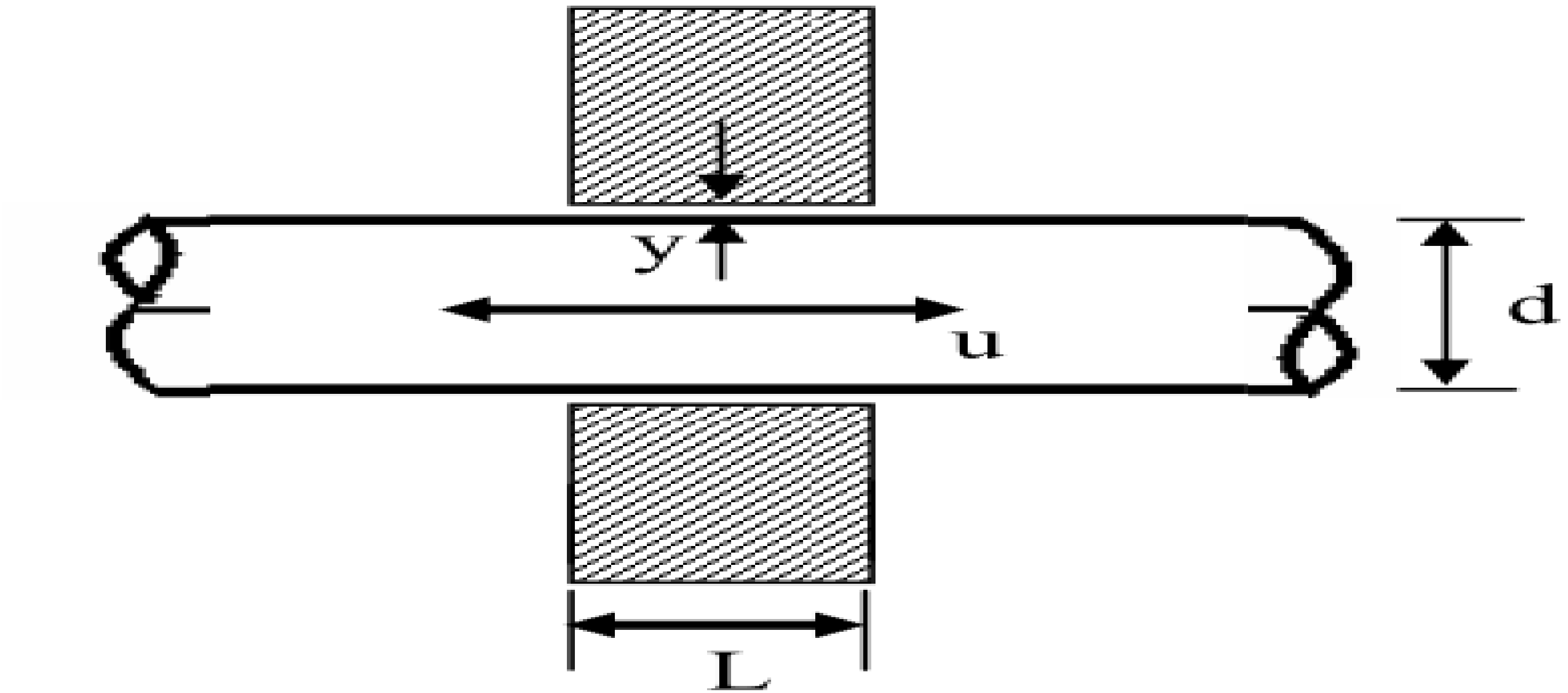
$$\tau = \mu \frac{du}{dy}$$

$Hence\ \tau = \frac{F}{A}$; ***So:*** $F = \mu A \frac{du}{dy}$

$$F = \mu A \frac{du}{dy}$$

Where: $\mathbf{A = \pi r^2 = \frac{\pi}{4} D^2}$ $\mathbf{m^2}$

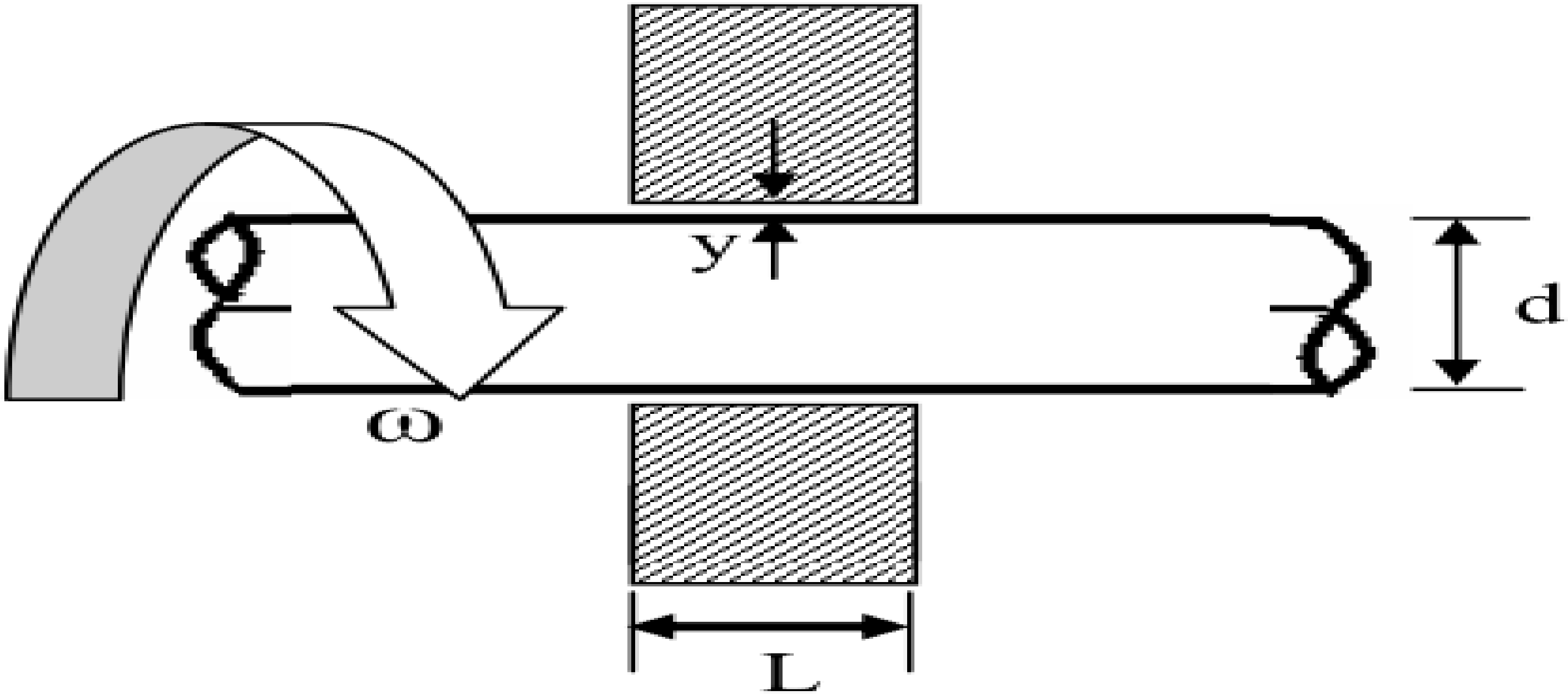
3. Moving Shaft



$$F = \mu A \frac{u}{y}$$

Where: $\mathbf{A = 2 \pi r L = \pi d L}$ $\mathbf{m^2}$

4. Rotating Shaft



$$F = \mu A \frac{u}{y}$$

Where: $u = \omega\, r = \frac{1}{2}\, \omega\, d$

$$\omega = \frac{2\pi N}{60}$$

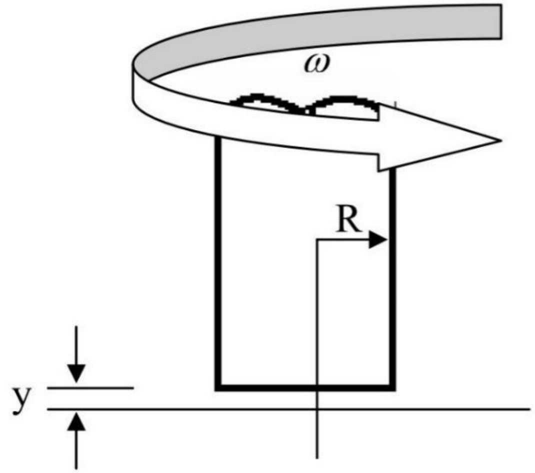
N in rpm and **ω** in rad/s

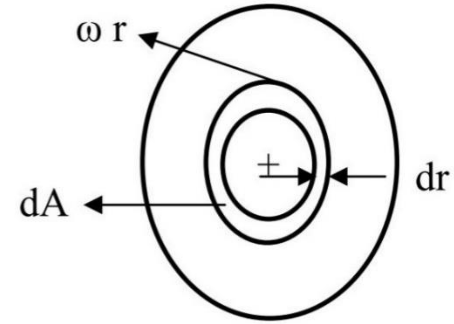
$$\tau = \mu \frac{du}{dy} \qquad \text{Hence } \tau = \frac{F}{A}; \qquad \textbf{So:} \quad F = \mu A \frac{du}{dy}$$

$$dA = 2\pi \quad rdr \quad \&$$

$$u = \omega r$$

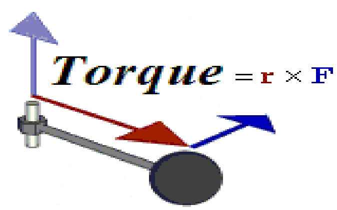
$$dF = \mu\,(2\pi\, rdr) * \frac{\omega r}{y}$$





$$F = \frac{\mu 2\pi\omega}{y} \int_0^R r^2\, dr \quad \Longrightarrow \quad F = \frac{2\pi\omega\mu}{3y} R^3$$

*Since the Torque =F*r or* $dT = rdF = r^* \ \mu\,(2\pi\, rdr) * \frac{\omega r}{y}$



$$\int_0^T dT = \frac{2\pi\omega\mu}{y} \int_0^R r^3\, dr \quad \Longrightarrow \quad T = \frac{2\pi\omega\mu}{4y} R^4$$

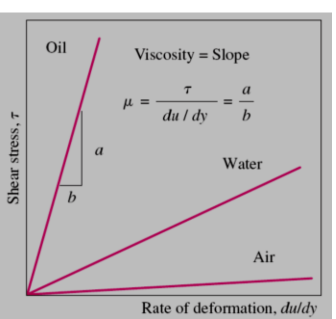
Linear variation of shearing stress with rate of shearing strain for common fluids

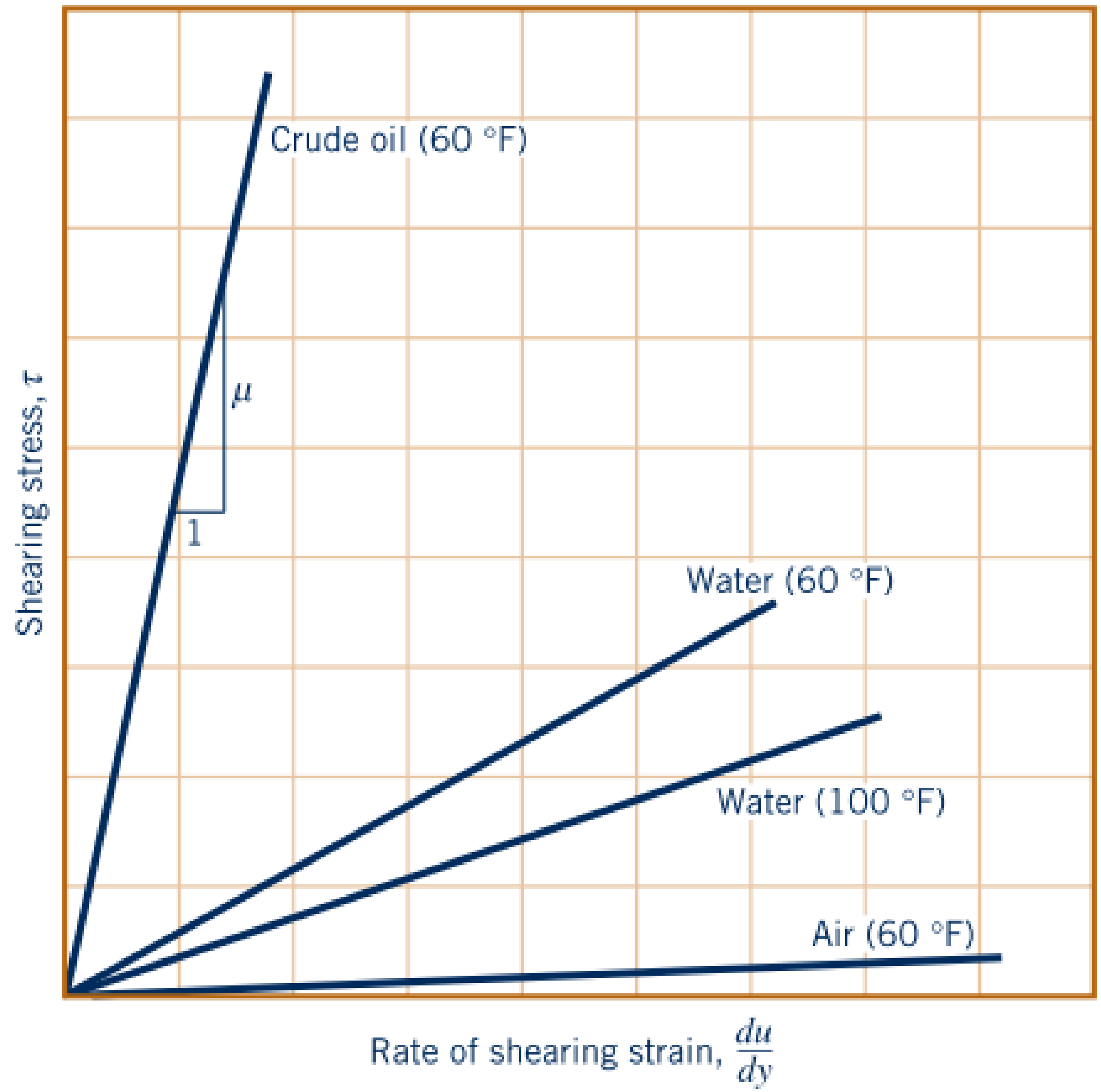
$$\tau \propto \frac{d(d\beta)}{dt}$$

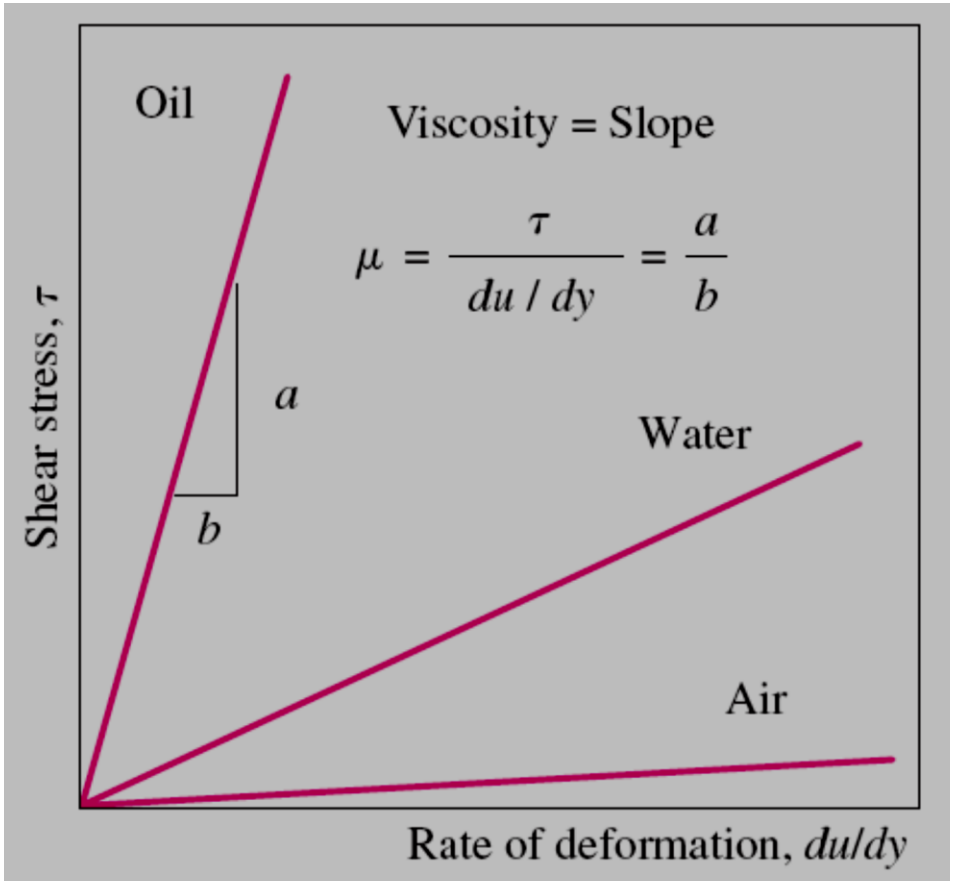
Or

$$\tau \propto \frac{du}{dy}$$

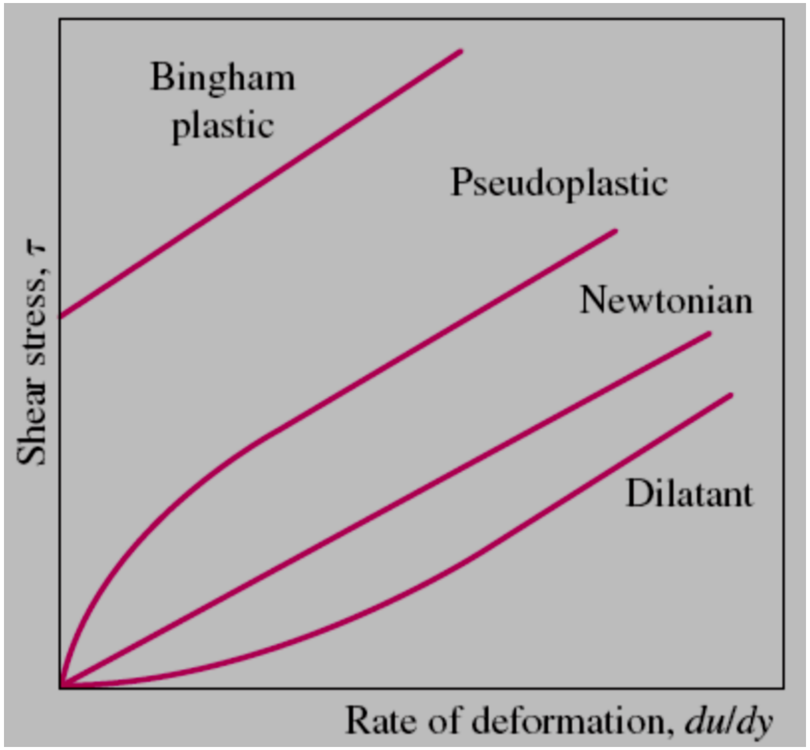
$$\tau = \mu \frac{du}{dy}$$







The rate of deformation (velocity gradient) of a Newtonian fluid is proportional to shear stress, and the constant of proportionality is the viscosity.



Variation of shear stress with the rate of deformation for Newtonian and non-Newtonian fluids (the slope of a curve at a point is the apparent viscosity of the fluid at that point).

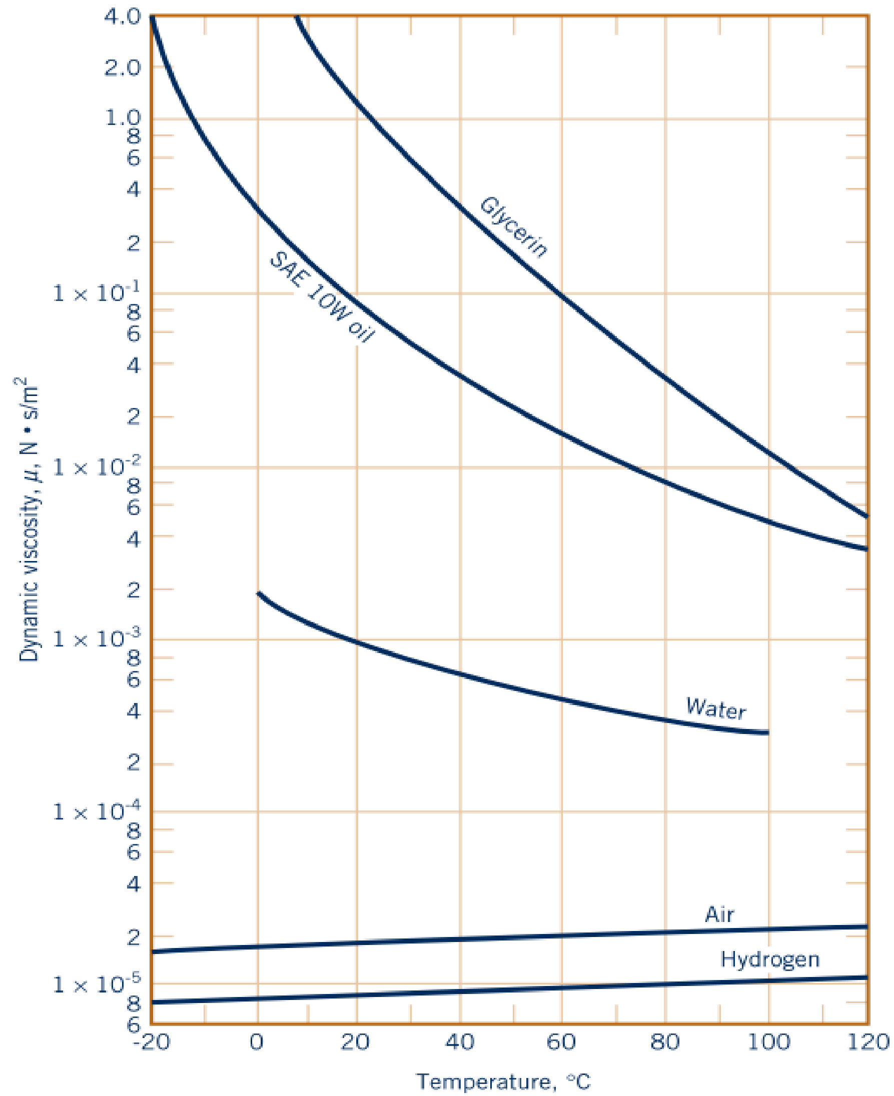
Viscosity & Temperature

It is to be noted from this figure that the **viscosity of liquids decreases** with an **increase in temperature**, whereas for **gases an increase** in **temperature causes an increase in viscosity.**

The effect of Temperature on Viscosity

viscosity of liquids ↓ with ↑ temperature,

viscosity of gases ↑ with ↑ temperature.



Dynamic (absolute) viscosity of some common fluids as a function of temperature.

Kinematic Viscosity

Kinematic viscosity (ν) is a fluid property obtained by dividing the dynamic viscosity (μ) by the fluid density

$\boldsymbol{\nu}$: is the Kinematic viscosity in m^2/s

$\boldsymbol{\mu}$: is the Dynamic viscosity in kg/m.s

$\boldsymbol{\rho}$: is the Density of a fluid in kg/m^3

$$\nu = \mu / \rho$$

Typical values:

Water = 1.14×10^{-6} m^2/s;

Air = 1.46×10^{-5} m^2/s;

Units of Viscosity

- The viscosity is the slope of the line of shear stress versus shear rate so its SI unit is 1 Pa / (1/s) = 1 Pa · s
- The customary unit of viscosity is the *poise* , *however* it is too large a unit for most common fluids.
- By sheer coincidence the viscosity of pure water at about is 0.01 poise; for that reason the common unit of viscosity is the centipoise.

$$1\,\mathrm{cP} = 0.01\frac{\mathrm{g}}{\mathrm{cm}\cdot\mathrm{s}}$$

$$1\,\mathrm{cP} = 0.001\,\mathrm{Pa}\cdot\mathrm{s}$$

$$1\,\mathrm{Pa}\cdot\mathrm{s} = 1\frac{\mathrm{kg}}{\mathrm{m}\cdot\mathrm{s}} \quad (= 1000\,\mathrm{cP})$$

$$1\,\mathrm{cP} = 6.72\times10^{-4}\ \frac{\mathrm{lbm}}{\mathrm{ft}\cdot\mathrm{s}}$$

Units of Kinematic Viscosity

$$\text{Kinematic viscosity} = \nu = \mu / \rho$$

The most common unit of kinematic viscosity is the centistoke (cSt)

$$1 \text{ cSt} = \frac{1 \text{ cP}}{1 \text{ g/cm}^3} = 10^{-6} \frac{\text{m}^2}{\text{s}} = 1.08 \times 10^{-5} \frac{\text{ft}^2}{\text{s}}$$

at 68°F = 20°C, water has a kinematic viscosity of $1.004 \approx 1$ cSt. To

Dynamic Viscosity and Kinematic Viscosity of Eight Fluids at 1 atm and 20°C

Fluid	μ, kg/(m · s)†	Ratio $\mu/\mu(H_2)$	ρ, kg/m^3	ν m^2/s†
Hydrogen	8.8 E–6	1.0	0.084	1.05 E–4
Air	1.8 E–5	2.1	1.20	1.51 E–5
Gasoline	2.9 E–4	33	680	4.22 E–7
Water	1.0 E–3	114	998	1.01 E–6
Ethyl alcohol	1.2 E–3	135	789	1.52 E–6
Mercury	1.5 E–3	170	13,580	1.16 E–7
SAE 30 oil	0.29	33,000	891	3.25 E–4
Glycerin	1.5	170,000	1,264	1.18 E–3

†1 kg/(m · s) = 0.0209 slug/(ft · s); 1 m^2/s = 10.76 ft^2/s.

Example

Suppose that the fluid being sheared as shown is SAE 30 oil at 20°C; μ = 0.29 kg/(m.s). Compute the shear stress in the oil if V = 3 m/s and h =2 cm.

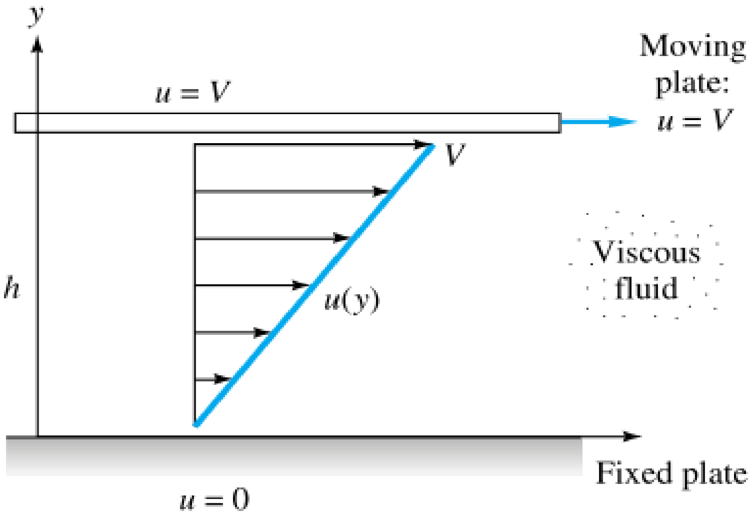
The shear stress is found from

$$\tau = \mu \frac{du}{dy} = \frac{\mu V}{h}$$

$$\tau = \frac{[0.29\ \text{kg/(m}\cdot\text{s)}](3\ \text{m/s})}{0.02\ \text{m}}$$

$$\tau = 43\ \text{kg/(m}\cdot\text{s}^2)$$

$$\tau = 43\ \text{kg/(m}\cdot\text{s}^2) = 43\ \text{N/m}^2 = 43\ \text{Pa}$$



Viscous flow induced by relative motion between two parallel plates.

Compressibility of Fluids Bulk Modulus

The compressibility, (change in volume due to change in pressure), of a liquid is inversely proportional to its volume modulus of elasticity; also known as the **bulk modulus**. This modulus that is commonly used to characterize compressibility is defined as:

Bulk modulus: $E_v \; or \; K = -\dfrac{d_P}{{d_V}/_V} = -V\dfrac{d_P}{d_V}$ N/m^2

Where: $\boldsymbol{d_P}$ is the differential change in pressure needed to create a differential change in volume.

So; the bulk modulus of elasticity of the fluid is the rate at which the pressure changes with the volumetric strain and is given in the above equation.

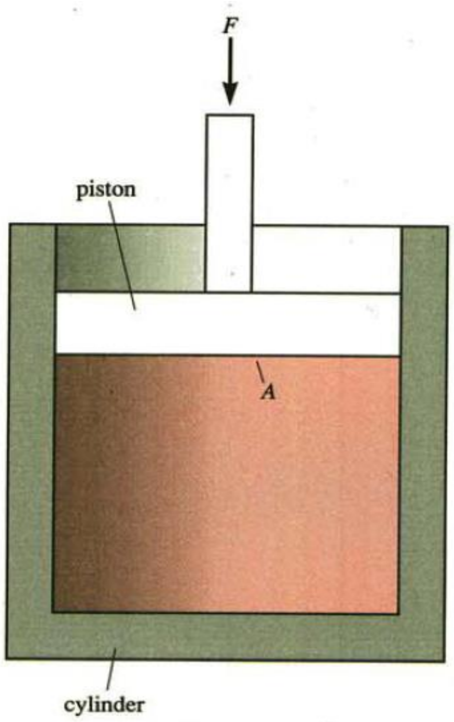
Compressibility of Fluids Bulk Modulus

Bulk modulus: E_v or $K = -\frac{d_P}{d_V/_V} = -V\frac{d_P}{d_V}$ N/m^2

The negative sign is there just to make K positive, since an increase in pressure will cause a decrease in volume. While the compressibility taken as β.

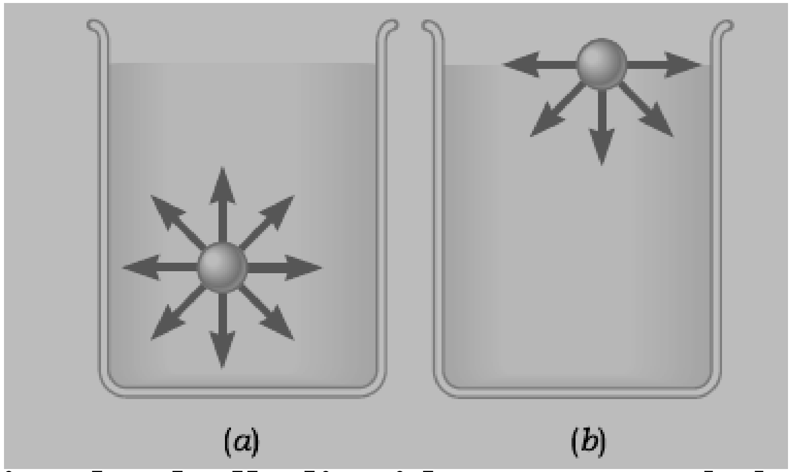
The quantity of compressibility is $(1/K)$

$$\beta = \frac{1}{K} = -{}^{1}\Big/_{V\frac{d_P}{d_V}}$$



Surface Tension

Surface tension is a property of liquids which is felt at the interface between the liquid and another fluid (typically a gas). Surface tension has dimensions of force per unit length (**N/m**), and always acts parallel to the interface. Surface molecules are subject to an attractive force from nearby surface molecules so that the surface is in a state of tension.

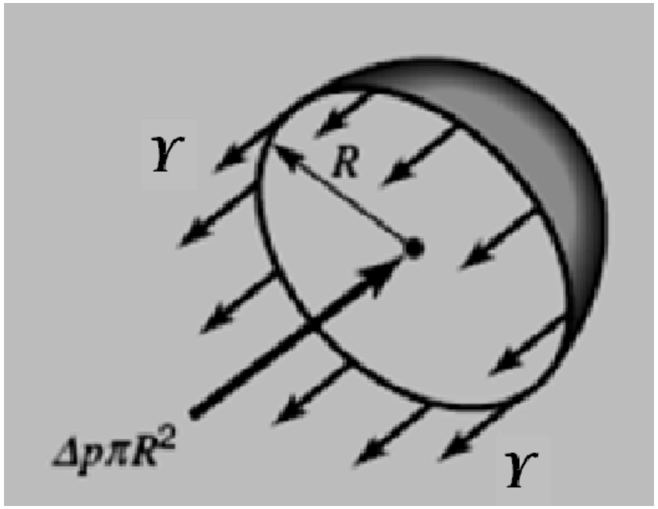


(a) A molecule **within the bulk liquid** is surrounded on all sides by other molecules, which attract it equally in all directions, leading to a zero net force.

(b) A molecule **in the surface** experiences a net attractive force pointing toward the liquid interior, because there are no molecules of the liquid above the surface

Surface Tension

The pressure inside a drop of fluid can be calculated using the free-body diagram If the spherical drop is cut in half (as shown) the force developed around the edge.



Forces acting on one-half of a liquid drop **2πRϒ**. This force must be balanced by the pressure difference between the internal and the external pressure **($\Delta P = P_i - P_e$)** that acting over the circular area, **πR^2**. Thus

$$2\pi R\Upsilon = \Delta_p \pi R^2 \quad \Rightarrow \quad \Delta_p = \frac{2\Upsilon}{R}$$

The Reynolds Number

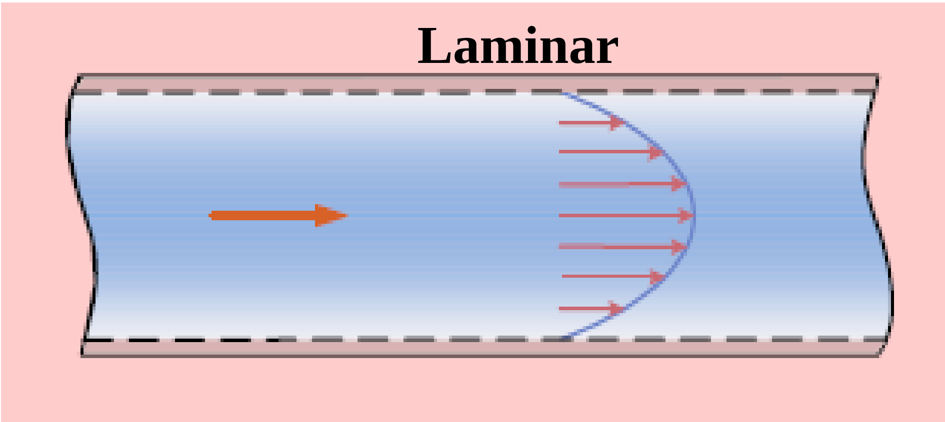
The primary parameter correlating the viscous behaviour of all Newtonian fluids is the dimensionless ***Reynolds number***

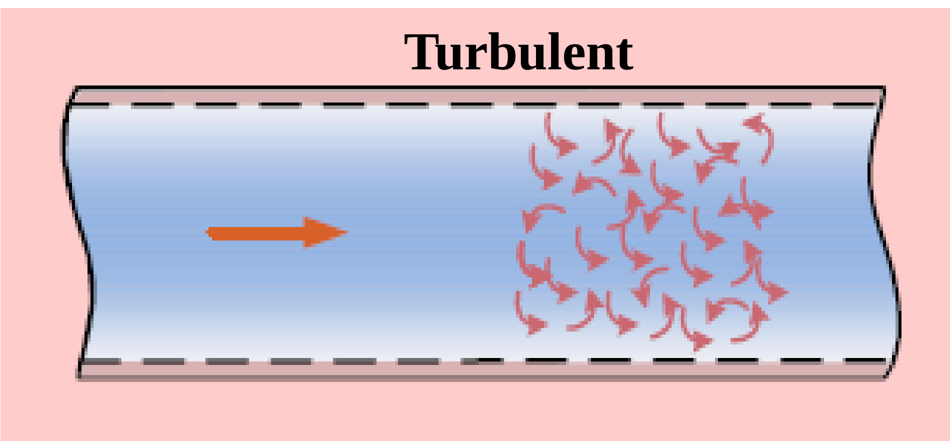
$$\mathrm{Re} = \frac{\rho V L}{\mu} = \frac{V L}{\nu}$$

Where:
V and L are characteristic velocity and length scales of the flow.

The second form of **Re** illustrates that the ratio of μ to ρ to has its own name, the kinematic viscosity ν.

The Reynolds Number





Generally, the first thing a fluids engineer should do is estimate the Reynolds number range of the flow under study.

Very low **Re** indicates viscous creeping motion, where inertia effects are negligible.

Moderate **Re** implies a smoothly varying laminar flow.

High **Re** probably spells turbulent flow, which is slowly varying in the time-mean but has superimposed strong random high frequency fluctuations.

Transition from laminar to turbulent flow occurs at a Re of 2200


Thank You